

DATA SCIENCE INTERN RESPONSE

1. Core KPIs (n=5,000 appointments)

Total appointments: 5,000

Unique patients: 1,696

Attendance rate: 46.3%

Overall no-show rate: 48.5%

Cancellation rate: 5.3%

These match your dashboard exactly.

2. Key analytical findings (denominator = Attended + No-Show only, n=4,737, since Cancelled appointments are excluded from this comparison)

No-show rate by previous no-shows: 0 → 46.3%, 1 → 55.9%, 2 → 62.1%, 3+ → 70.3%

No-show rate, 31-60 day lead time: 63.9%

No-show rate, reminder sent vs not: 49.9% (sent) vs 54.6% (not sent)

No-show rate, distance ≥20km: 60.2%

These are very close to your original figures (43.5/53.5/59.4/68.8, 60.5%, 47.4%/51.4%, ~58%) small differences likely come from slightly different denominators or rounding, but the patterns match closely. Good cross-validation.

3. Segment validation does the 31-60 day pattern hold across appointment types? Yes, consistently elevated across all four types:

Follow-up: 68.8% (highest)

Specialist Consultation: 63.0%

Diagnostic Test: 62.5%

General Consultation: 61.4%

No appointment type stands out as an exception the 31-60 day risk pattern is not concentrated in one type, it's broad-based.

4. Data Science Week 7 findings relevant to these variables:

Confirmed my model's top-risk predictions align 98.3% with your defined high-risk segments (3+ previous no-shows, 31-60 day lead time, 20km+ distance) strong cross-validation of both our approaches.

Important finding to flag: I discovered a gender fairness gap in my model (ROC-AUC 0.725 for male patients vs 0.639 for female) that none of the variables above directly explain. I tried adding gender-interaction features but it didn't close the gap. If you have bandwidth, it might be worth checking whether any of these 4 variables (previous no-shows, lead time, reminders, distance) show a different pattern by gender in your dashboard that could help explain what my model is missing.

Let me know if you need anything else for your documentation!